

BUILDTRACK

Construction Project Management & Site Monitoring Platform

MILESTONE 3 DOCUMENTATION

Modules 5–7: Material Inventory, Workforce Management & Procurement Management

Document Item	Details
Project	BuildTrack – Construction Project Management & Site Monitoring Platform
Milestone	Milestone 3 Documentation
Modules Covered	Modules 5–7: Material Inventory, Workforce Management & Procurement Management
Technology Evidence	Angular frontend; Python FastAPI backend; SQLAlchemy/database layer
Document Type	Technical Project / Internship Documentation
Evidence Basis	Current BuildTrack project source and reference milestone specification
Version	1.0

1. Milestone Overview

Milestone 3 documents the material/inventory, workforce and procurement areas of BuildTrack. The source project contains a Material Inventory placeholder screen, a detailed Workforce Management workspace, and a detailed Procurement Management workspace. The reference specification identifies material procurement, inventory monitoring, material requests, material allocation, stock management, workforce operations and procurement operations as the intended scope. Only functionality evidenced in the current project is described as implemented.

Module	Purpose	Main Implemented / Observed Areas
Module 5 – Material Inventory	Provide a material/inventory area for construction materials.	Current source contains an Admin Material Inventory placeholder page; full inventory workflows are not evidenced in the current source.
Module 6 – Workforce Management	Manage construction workers, attendance, allocation, shifts and payroll.	Dashboard, Worker Registration, Attendance Tracking, Workforce Allocation, Shift Scheduling, Payroll Monitoring, Workforce Categories and Workforce Reports.
Module 7 – Procurement Management	Manage procurement requests, vendors/suppliers, purchase orders and invoices.	Dashboard, Procurement Requests, Purchase Orders, Invoice Tracking, Vendor Management, Supplier Management, Procurement Categories and Procurement Reports.

3. Milestone Objectives

- Provide a dedicated Material Inventory entry point while clearly recording any incomplete inventory functionality.
- Provide workforce visibility through worker registration, attendance, allocation, shift scheduling, categories, payroll and reports.
- Provide procurement workflows for requests, vendors, suppliers, purchase orders, invoices, categories and reports.
- Present construction operations through structured Angular pages, forms, tables, search fields and dashboard summaries.
- Maintain a clear distinction between implemented frontend UI and backend persistence that requires verification.

2. System Architecture Used in This Milestone

BuildTrack uses an Angular frontend with route-based page components and a Python FastAPI backend. The project also contains a SQLAlchemy/database layer. The documentation distinguishes visible frontend interfaces from backend persistence so that a page is not automatically treated as proof of a live API/database workflow.

Layer	Observed Implementation
Frontend	Angular project structure with standalone components, templates, forms, tables and dashboard-style interfaces.

Routing	Dedicated routes/pages exist for Material Inventory, Workforce Management, Procurement Management, Notifications, Dashboard & Analytics, Reports and Budget Management.
Backend	Python FastAPI application with multiple routers and database-session integration.
Database	SQLAlchemy engine/session configuration and ORM/database layer are present.
Authorization	authGuard and roleGuard are defined; protected module routes should be described according to the actual route configuration.

4. Module 5 – Material Inventory

4.1 Objective

The Material Inventory module is intended to support material procurement, inventory monitoring, material requests, material allocation and stock management. In the current BuildTrack source inspected for this documentation, the Admin Material Inventory area contains a Material Inventory page, but the available implementation is only a placeholder. Therefore, the full reference feature set is not claimed as implemented.

4.2 Current BuildTrack Implementation Status

Area	Observed Status
Material Inventory page	A dedicated Admin Material Inventory page exists in the Angular source.
Inventory functionality	The current page is a placeholder rather than a demonstrated end-to-end inventory workflow.
Material Procurement	Not specified as implemented in the inspected current source.
Inventory Monitoring	Not specified as implemented in the inspected current source.
Material Requests	Not specified as implemented in the inspected current source.
Material Allocation	Not specified as implemented in the inspected current source.
Stock Management	Not specified as implemented in the inspected current source.

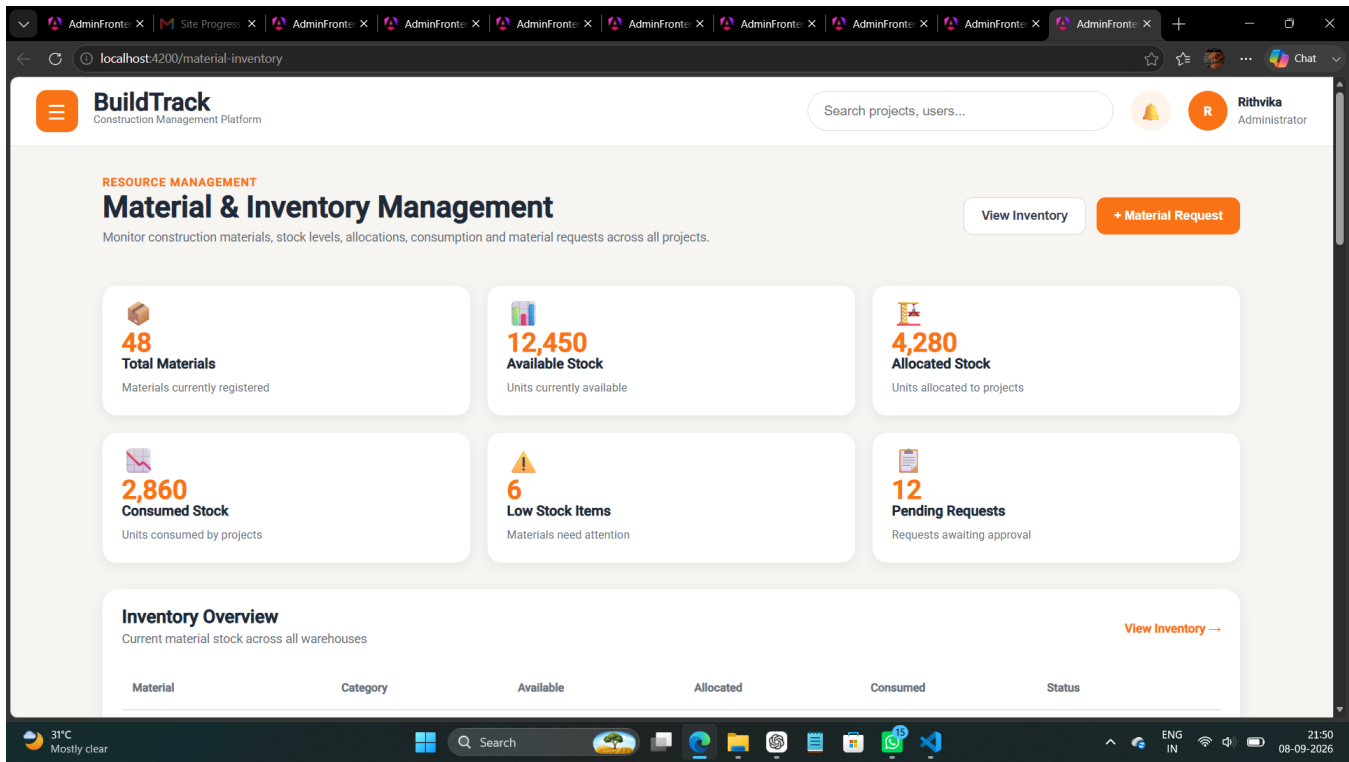


Figure 1: BuildTrack – Material Inventory Page

Screenshot note: capture the actual Material Inventory screen from the latest running version. Do not label placeholder content as a completed inventory workflow.

4.3 Inventory Management

The Inventory Management page allows users to monitor and manage materials available for construction projects.

It provides information about material names, categories, quantities and current stock levels.

This helps ensure that required materials are available for ongoing project activities.

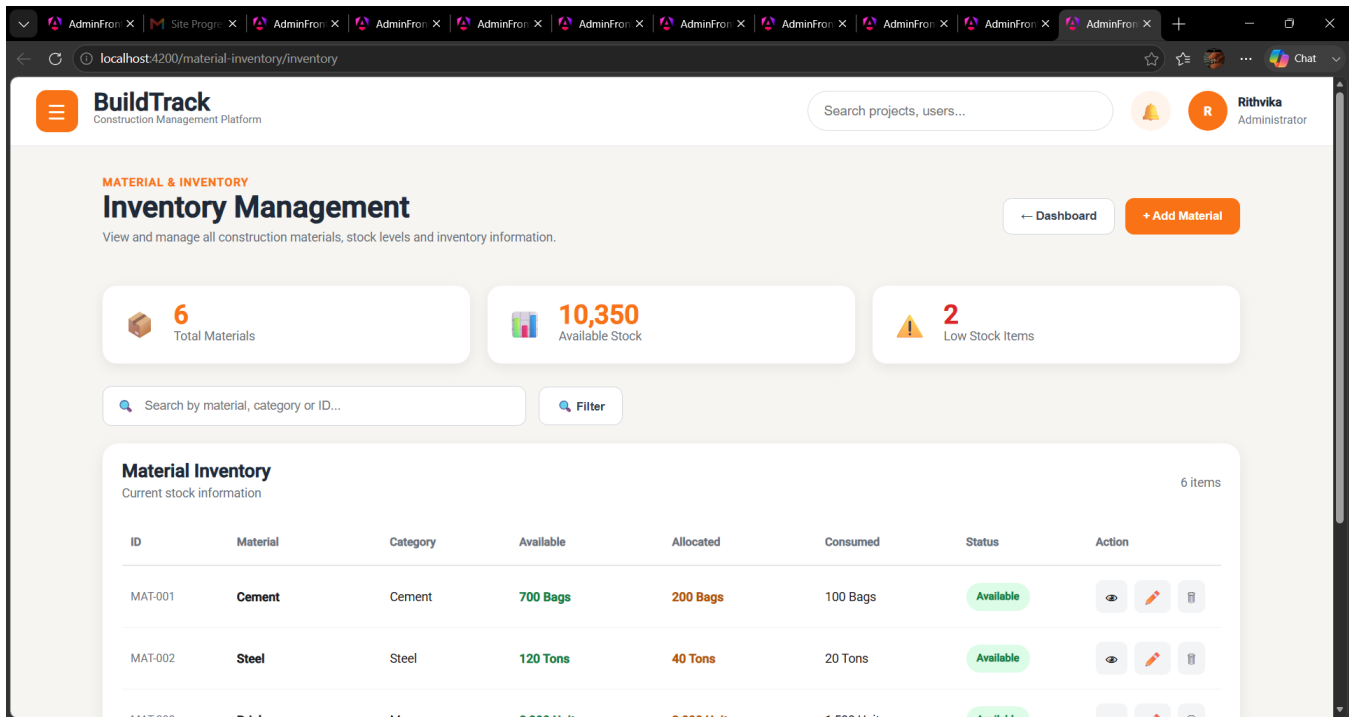


Figure 2: BuildTrack – Inventory Management Page

4.4 Material Request

The Material Request page allows users to view and monitor requests raised for construction materials. It provides information about requested materials, quantities, projects and request status. This helps track material requirements and supports timely procurement and allocation.

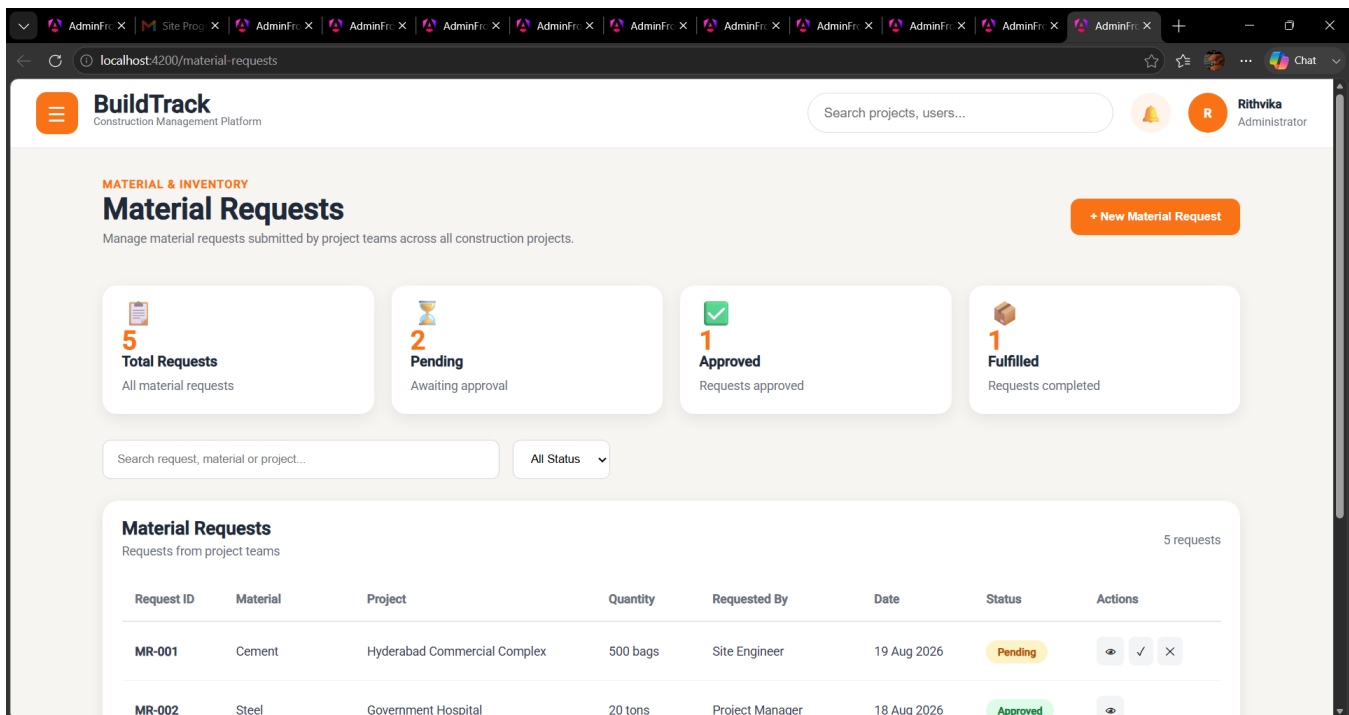
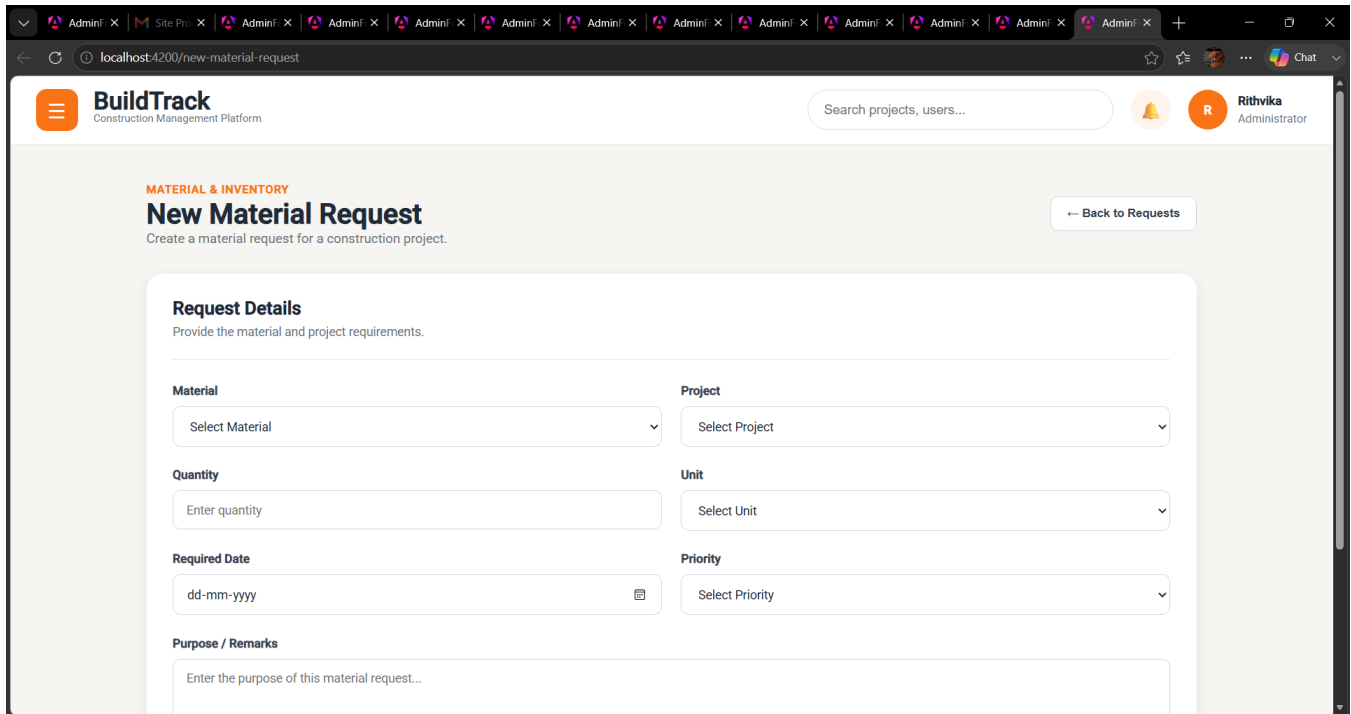


Figure 3: BuildTrack – Material request Page

4.5 New Material Request

The New Material Request page allows users to create a request for required construction materials. The user can provide details such as the material, required quantity, project and request information. After submission, the request can be reviewed and processed for further material procurement.



The screenshot displays the 'New Material Request' page in the BuildTrack application. The page header includes the BuildTrack logo, a search bar, and a user profile for 'Rithvika Administrator'. The main heading is 'New Material Request' with a subtitle 'Create a material request for a construction project.' and a 'Back to Requests' button. The form is titled 'Request Details' and contains the following fields:

- Material:** A dropdown menu with 'Select Material'.
- Project:** A dropdown menu with 'Select Project'.
- Quantity:** A text input field with 'Enter quantity'.
- Unit:** A dropdown menu with 'Select Unit'.
- Required Date:** A date input field with 'dd-mm-yyyy' and a calendar icon.
- Priority:** A dropdown menu with 'Select Priority'.
- Purpose / Remarks:** A text area with 'Enter the purpose of this material request...'.

Figure 4: BuildTrack – New Material request Page

4.6 Low Stock

The Low Stock page helps users identify materials whose available quantity has reached a low stock level. It provides information about materials that may require replenishment. This helps prevent material shortages and supports continuous construction activities.

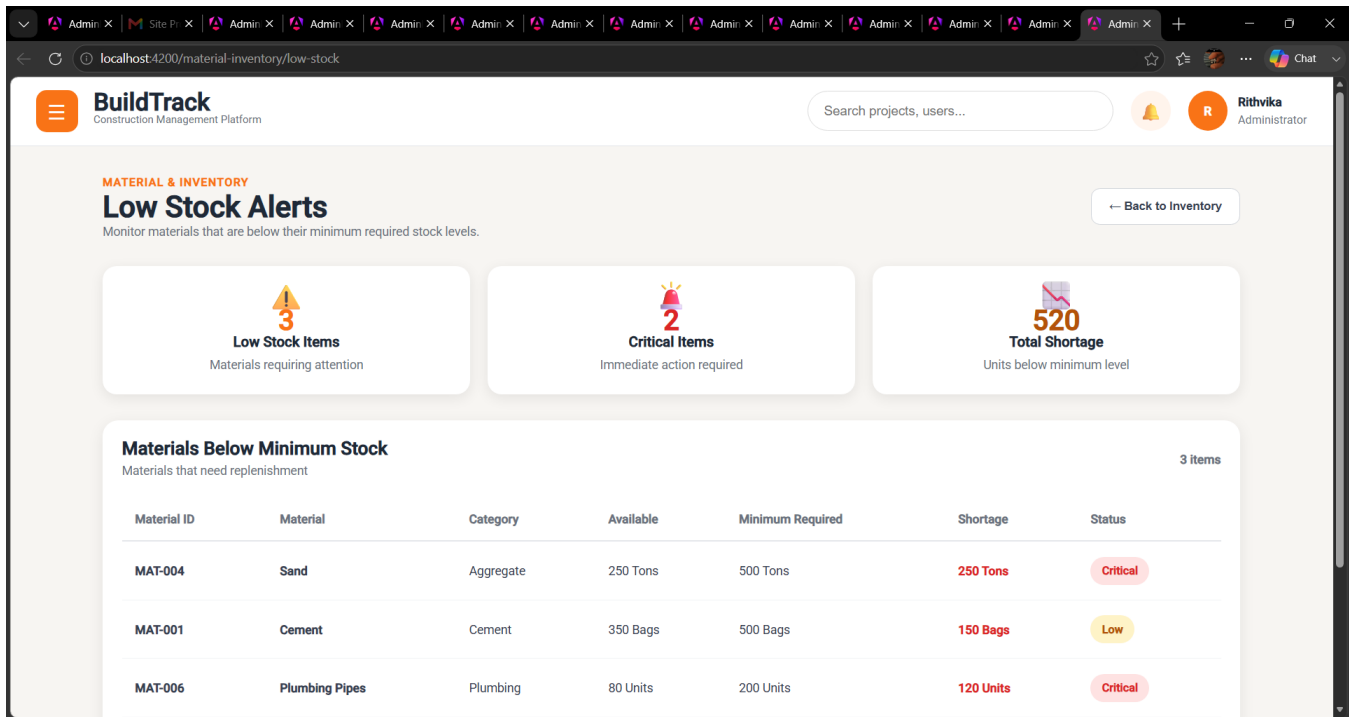
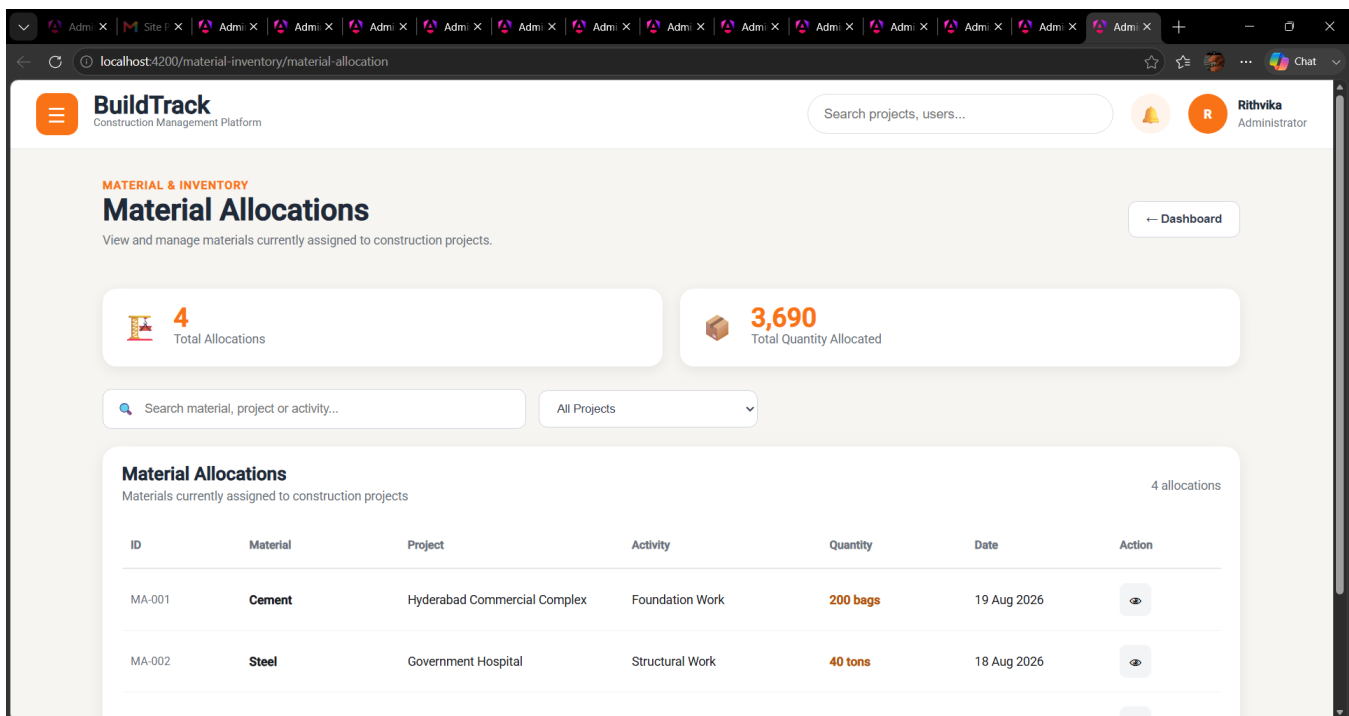


Figure 5: BuildTrack – Low stock Page

4.7 Material Allocation

The Material Allocation page allows users to allocate available materials to specific projects or construction activities. It provides information about the material, allocated quantity and associated project or work requirement. This helps ensure that materials are distributed according to project requirements.

Figure 6: BuildTrack – Material allocation Page



4.8 Stock Movements

The Stock Movements page allows users to monitor changes in material stock levels.

It provides information about material movements such as stock additions, issues and other inventory changes.

This helps maintain visibility of material flow and supports accurate inventory monitoring.

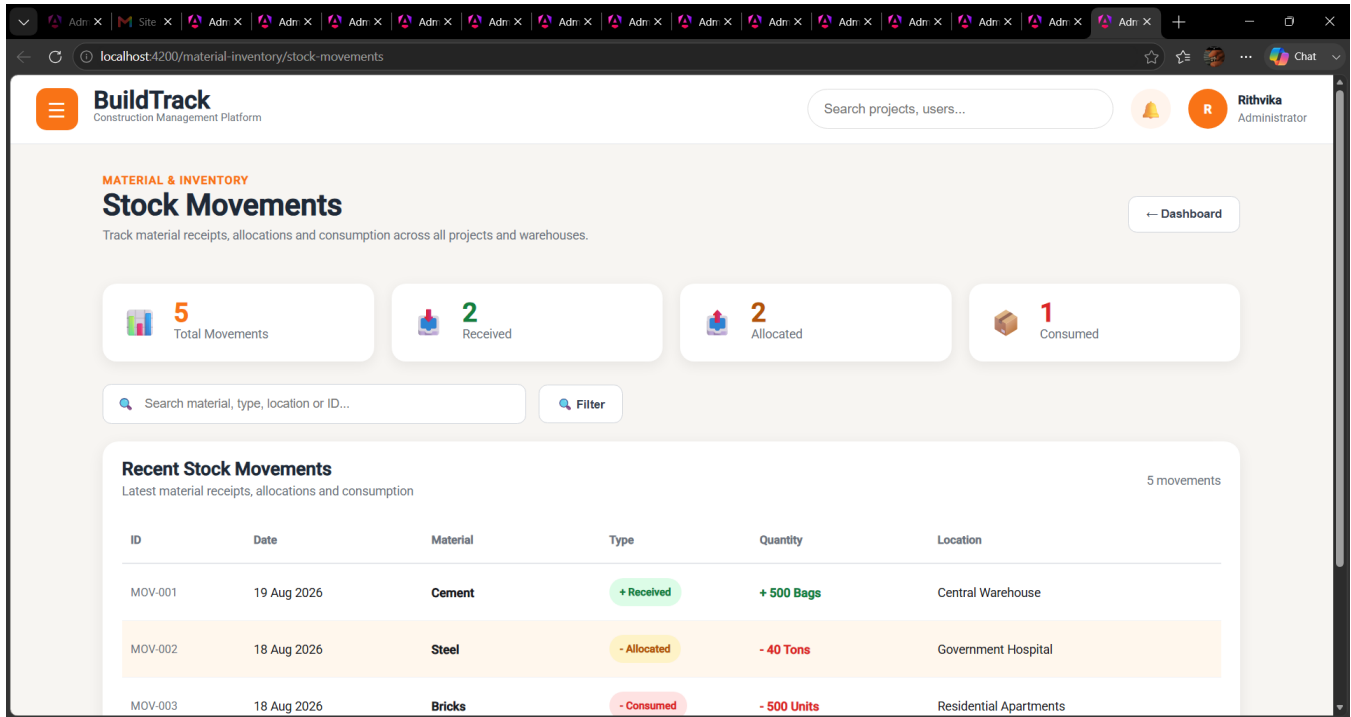


Figure 7: BuildTrack – Stock movements Page

Module Outcome

The Material Inventory module is represented by a dedicated frontend entry point, but the inspected implementation does not provide sufficient evidence to claim the complete inventory feature set. The module should be marked as partial/pending until its actual workflows and persistence are demonstrated.

5. Module 6 – Workforce Management

5.1 Objective

The Workforce Management module provides a structured workspace for registering workers, monitoring attendance, allocating workforce to projects, scheduling shifts, managing workforce categories, monitoring payroll and generating workforce reports.

5.2 Workforce Dashboard

Purpose: Provide a consolidated workforce overview.

How it works in BuildTrack: The dashboard displays total workers, present workers, absent workers, workers on leave, active assignments, active shifts, pending payroll and an attendance percentage. It also includes workforce categories, attendance overview and project-wise workforce sections.

Role/User: Workforce Management user

Expected output: A summarized workforce status and project-wise workforce view.

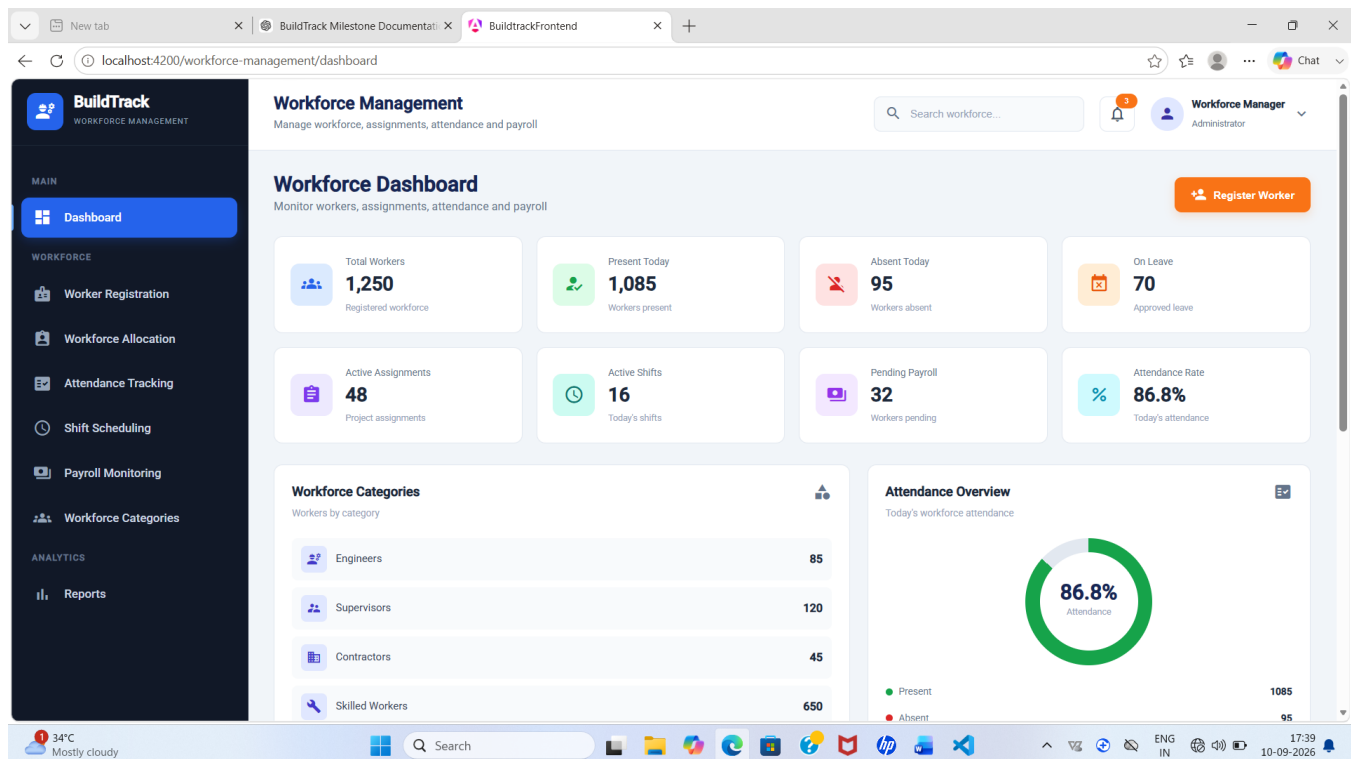


Figure 2: BuildTrack – Workforce Dashboard

5.3 Worker Registration

Purpose: Register and maintain worker information.

How it works in BuildTrack: The page provides worker registration fields such as Worker ID, worker name, contact information and skill/work type. It also provides active/leave/inactive summary information, a registered-worker table, search, and a bulk workforce-file upload area.

Role/User: Workforce Management user

Expected output: Worker records can be entered/reviewed through the registration interface.

Worker Registration
Register and maintain workforce information for construction projects.

Registered Workers
Manage registered workforce information.

WORKER ID	WORKER	CATEGORY	SKILL / WORK TYPE	CONTRACTOR	PROJECT	JOINING DATE	STATUS
WRK-1001	Arun Kumar 9876543210	Skilled Workers	Masonry	ABC Constructions	Chennai Metro Construction	2026-01-15	Active
WRK-1002	Ravi Shankar 9876501234	Engineers	Civil Engineering	Prime Infra	Airport Expansion	2026-02-10	Active
WRK-1003	Suresh B 9988776655	Unskilled Workers	General Labour	BuildPro Contractors	Green Valley Apartments	2026-03-05	Active

Figure 3: BuildTrack – Worker Registration

5.4 Attendance Tracking

Purpose: Monitor workforce attendance records.

How it works in BuildTrack: The page summarizes total workers, present, absent and late counts and provides an Attendance Records table with Attendance ID, Worker, Project, Date, Shift, Check In, Check Out and Status.

Role/User: Workforce Management user

Expected output: Attendance status and attendance records are presented for monitoring.

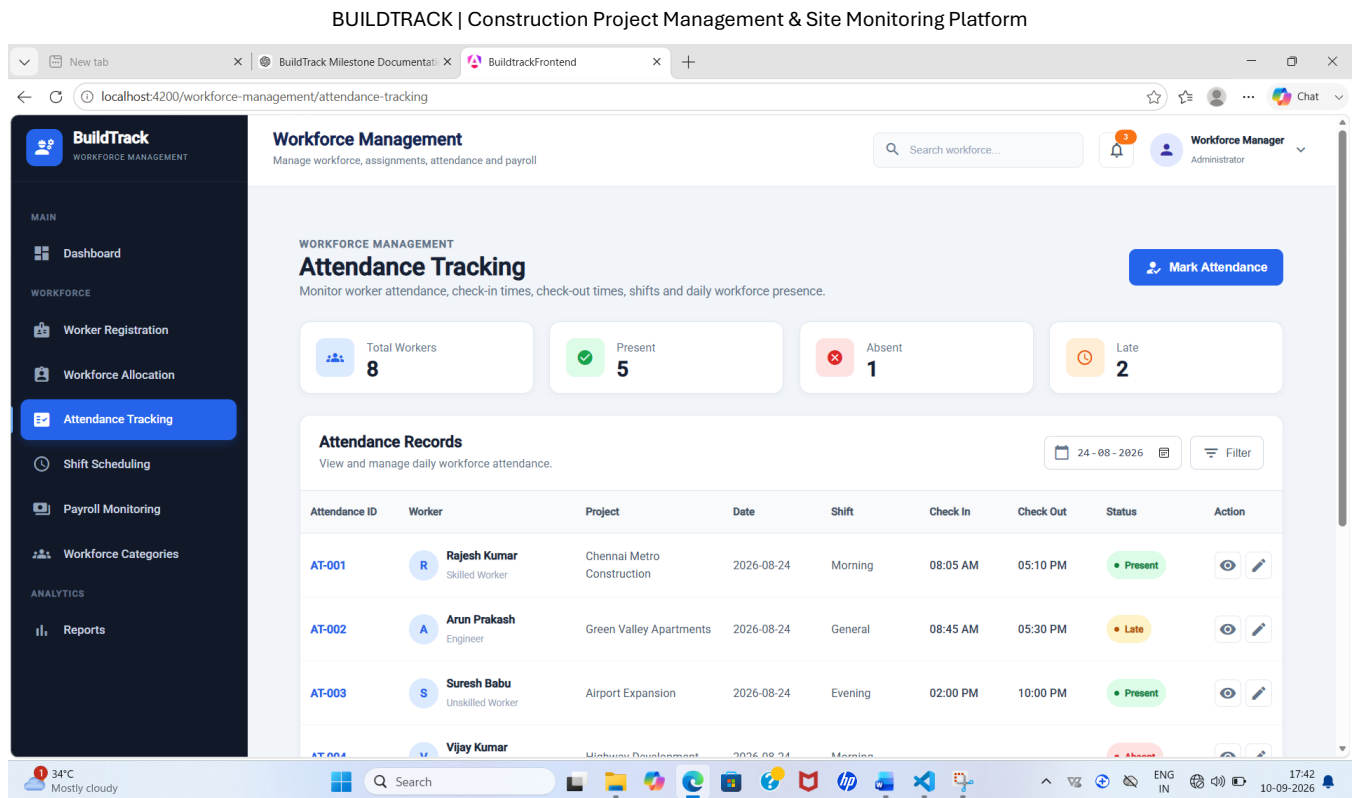


Figure 4: BuildTrack – Attendance Tracking

5.5 Workforce Allocation

Purpose: Assign workers to construction projects and work areas.

How it works in BuildTrack: The page provides allocation summary values and a New Workforce Allocation area. The current allocation table includes Allocation ID, Worker, Category, Project, Work Area, Contractor, Shift, Start Date, End Date and Status.

Role/User: Workforce Management user

Expected output: Workforce allocation information is available for project-level monitoring.

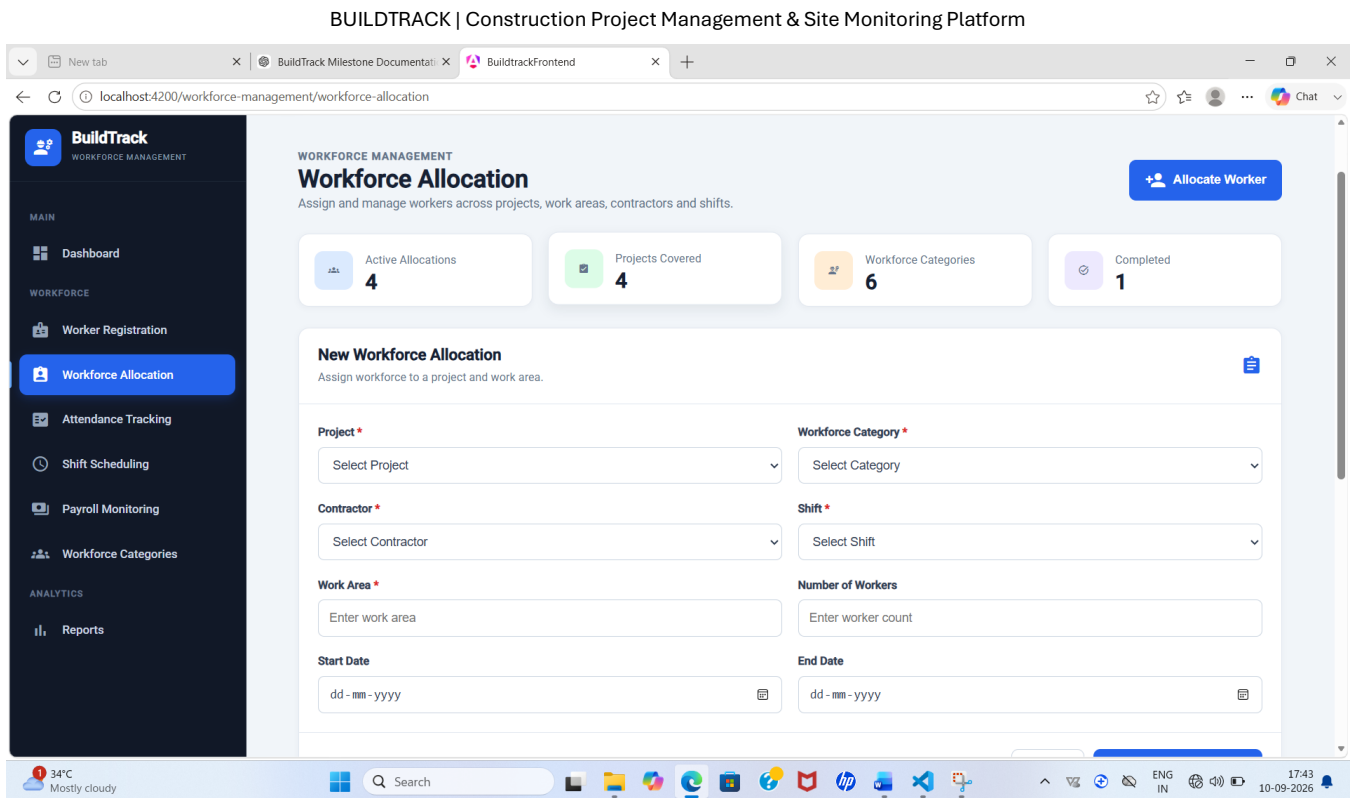


Figure 5: BuildTrack – Workforce Allocation

5.6 Shift Scheduling

Purpose: Plan and monitor workforce shifts.

How it works in BuildTrack: The page summarizes total schedules, morning shifts, active shifts and night shifts. A Create New Shift Schedule area and a Current Shift Schedules table are provided with project, category, work area, shift, date, workers, supervisor, timing and status.

Role/User: Workforce Management user

Expected output: Scheduled workforce shifts can be reviewed through the page.

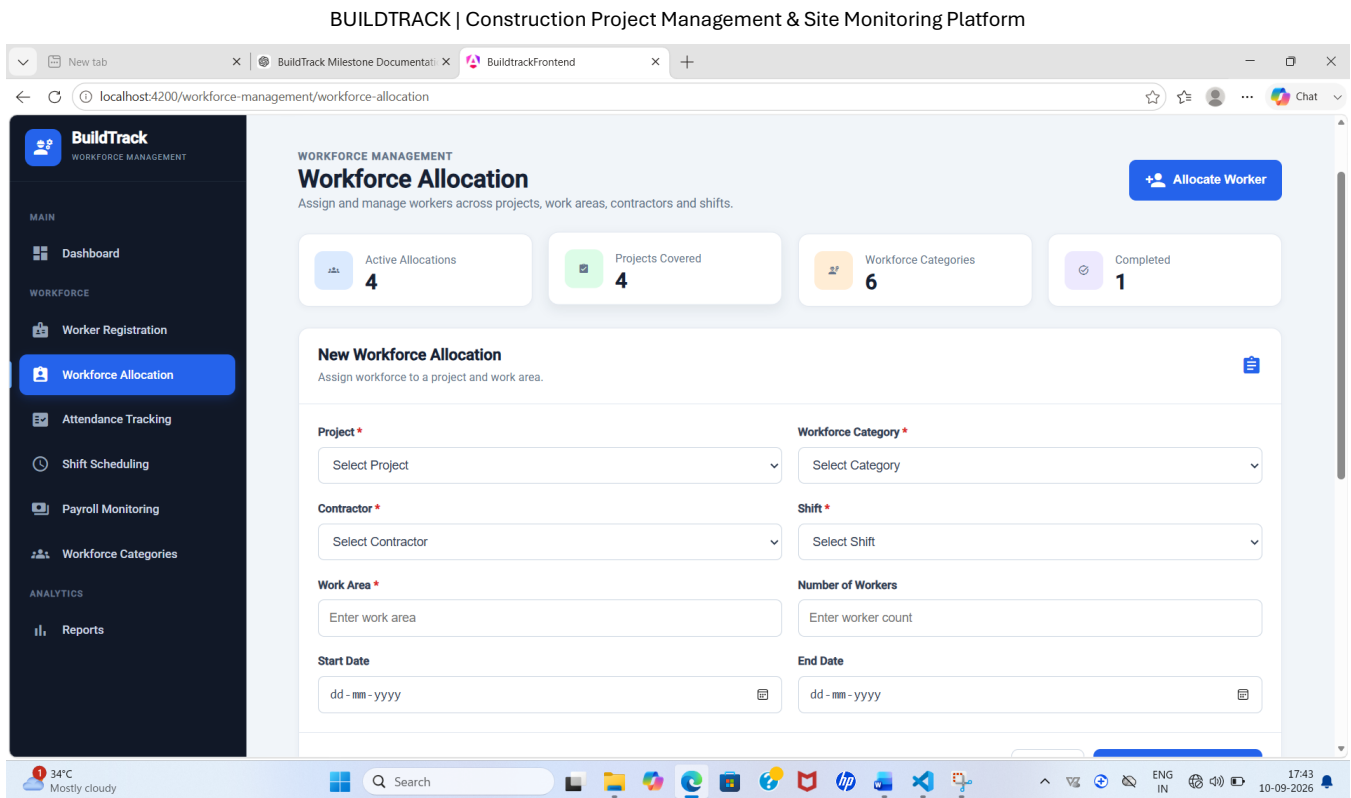


Figure 6: BuildTrack – Shift Scheduling

5.7 Payroll Monitoring

Purpose: Monitor workforce payroll information.

How it works in BuildTrack: The page displays total payroll, processed payroll, pending payroll and worker count. The Payroll Monitoring table includes Payroll ID, Worker, Project, Category, Pay Period, Basic Pay, Overtime, Deductions, Net Pay and Status.

Role/User: Workforce Management user

Expected output: Payroll processing and monitoring information is presented.

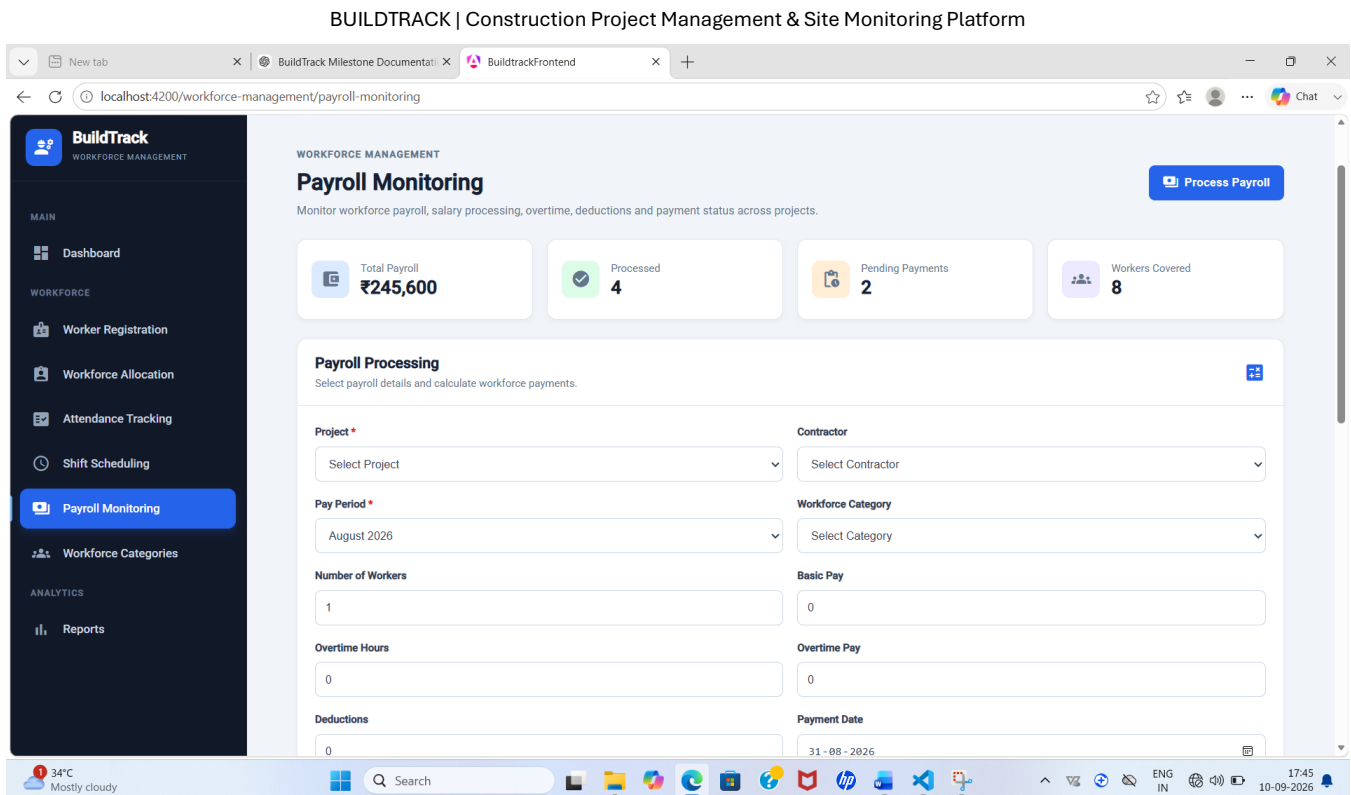


Figure 7: BuildTrack – Payroll Monitoring

5.8 Workforce Categories

Purpose: Organize workers by workforce category.

How it works in BuildTrack: The page provides category summary values and a Workforce Category Setup/List area for managing category-related information.

Role/User: Workforce Management user

Expected output: Workforce categories can be organized and reviewed.

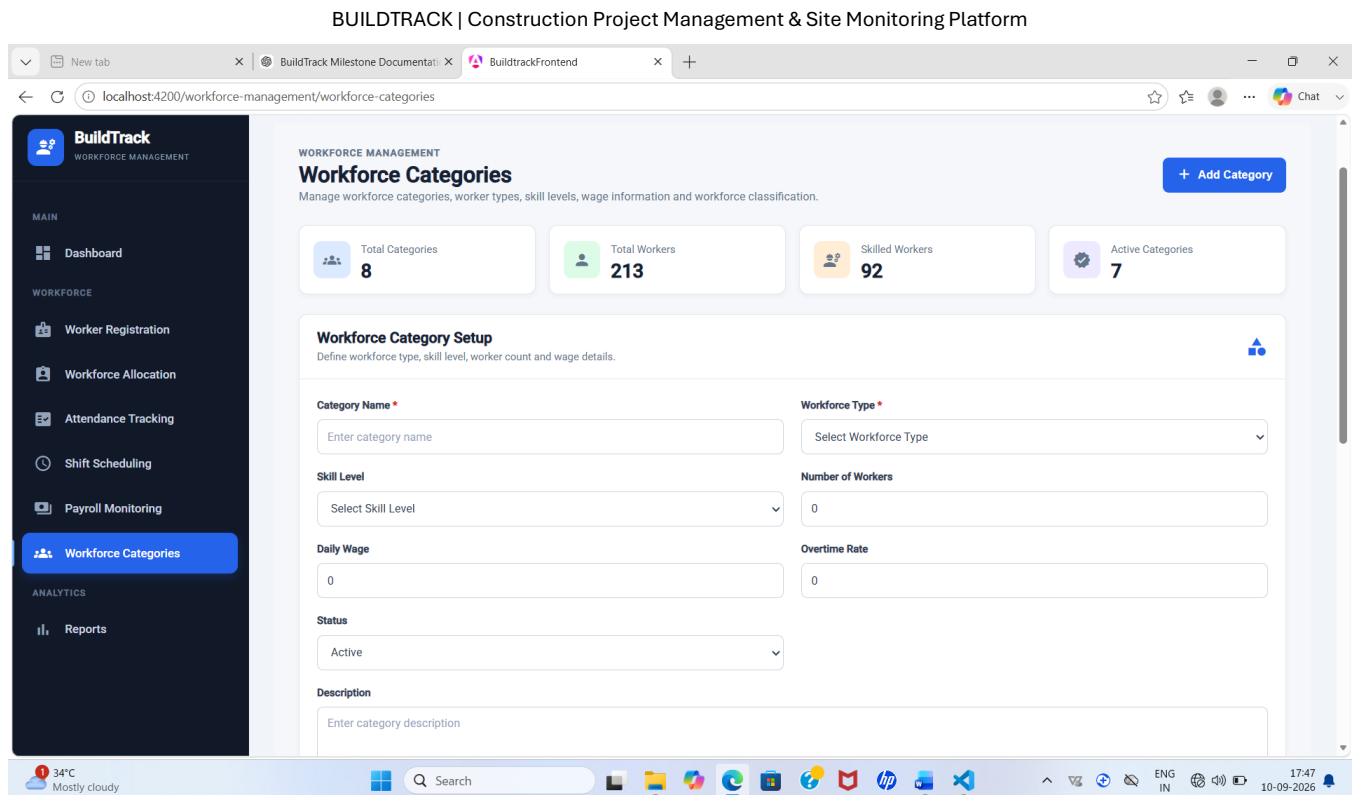


Figure 8: BuildTrack – Workforce Categories

5.9 Workforce Reports

Purpose: Provide workforce-focused reporting.

How it works in BuildTrack: The Workforce Reports page contains a Report Generator and Workforce Report Overview area for presenting workforce reporting information.

Role/User: Workforce Management user

Expected output: Workforce reporting information can be accessed.

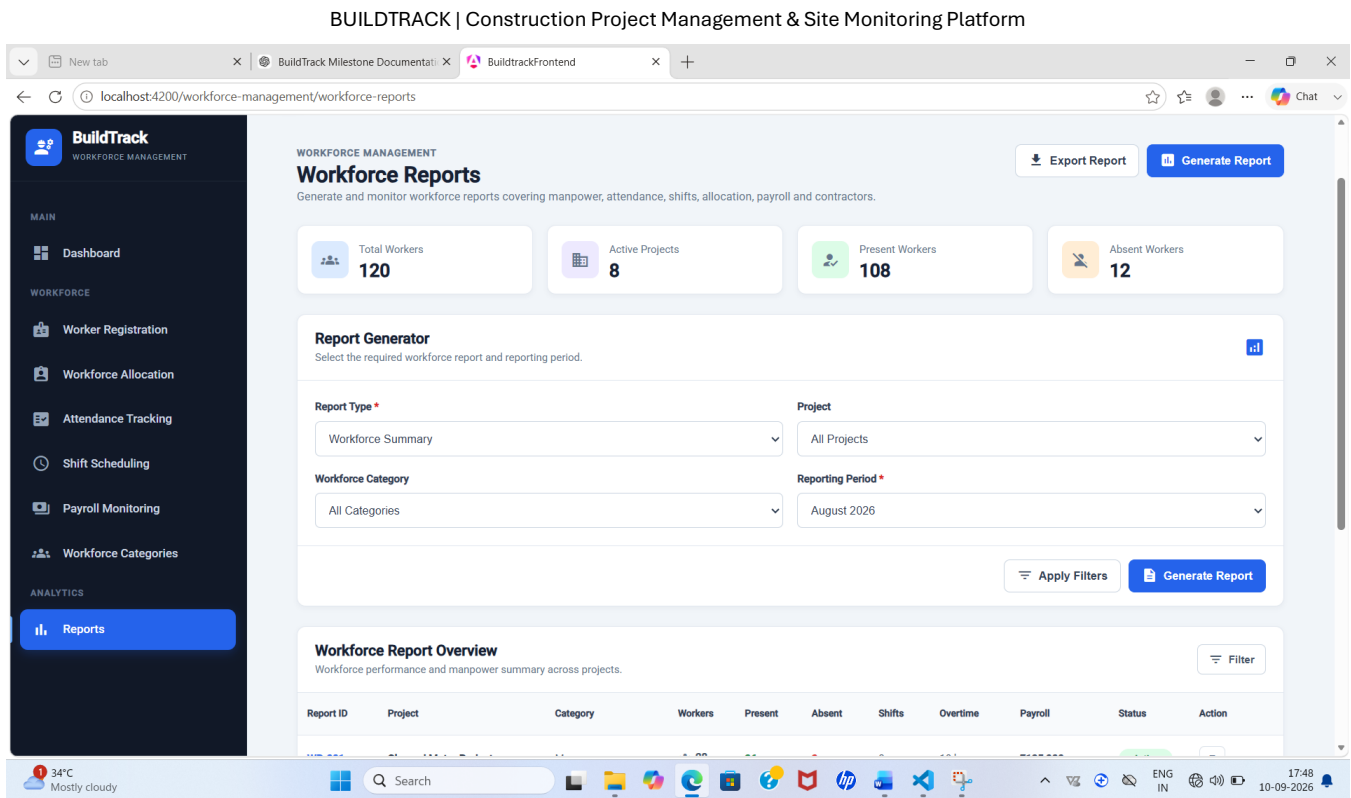


Figure 9: BuildTrack – Workforce Reports

End-to-End Workflow

Step	User Action	System Response	Result
1	Open Workforce Dashboard	Dashboard loads workforce summary indicators and project-wise workforce information.	Overall workforce status is available.
2	Register or review workers	Worker Registration displays worker fields, status summaries and the registered-worker table.	Worker information can be maintained/reviewed.
3	Review attendance	Attendance Tracking displays attendance counts and detailed attendance records.	Attendance status is available for monitoring.
4	Allocate workforce	Workforce Allocation presents worker/project/work-area/shift allocation information.	Workforce assignments can be reviewed.
5	Schedule shifts	Shift Scheduling displays schedule summaries and current shift schedules.	Shift planning information is available.
6	Review payroll	Payroll Monitoring presents payroll totals and payroll records.	Payroll status can be monitored.

Module Outcome

The Workforce Management module provides dedicated frontend interfaces covering the main operational areas of workforce registration, attendance, allocation, shift scheduling, categories, payroll and reporting.

6. Module 7 – Procurement Management

6.1 Objective

The Procurement Management module provides a structured procurement workspace for managing procurement requests, vendors, suppliers, purchase orders, invoices, procurement categories and procurement reports.

6.2 Procurement Dashboard

Purpose: Provide a centralized procurement overview.

How it works in BuildTrack: The page presents a Procurement Dashboard with summary information and procurement navigation.

Role/User: Procurement Management user

Expected output: High-level procurement information is presented.

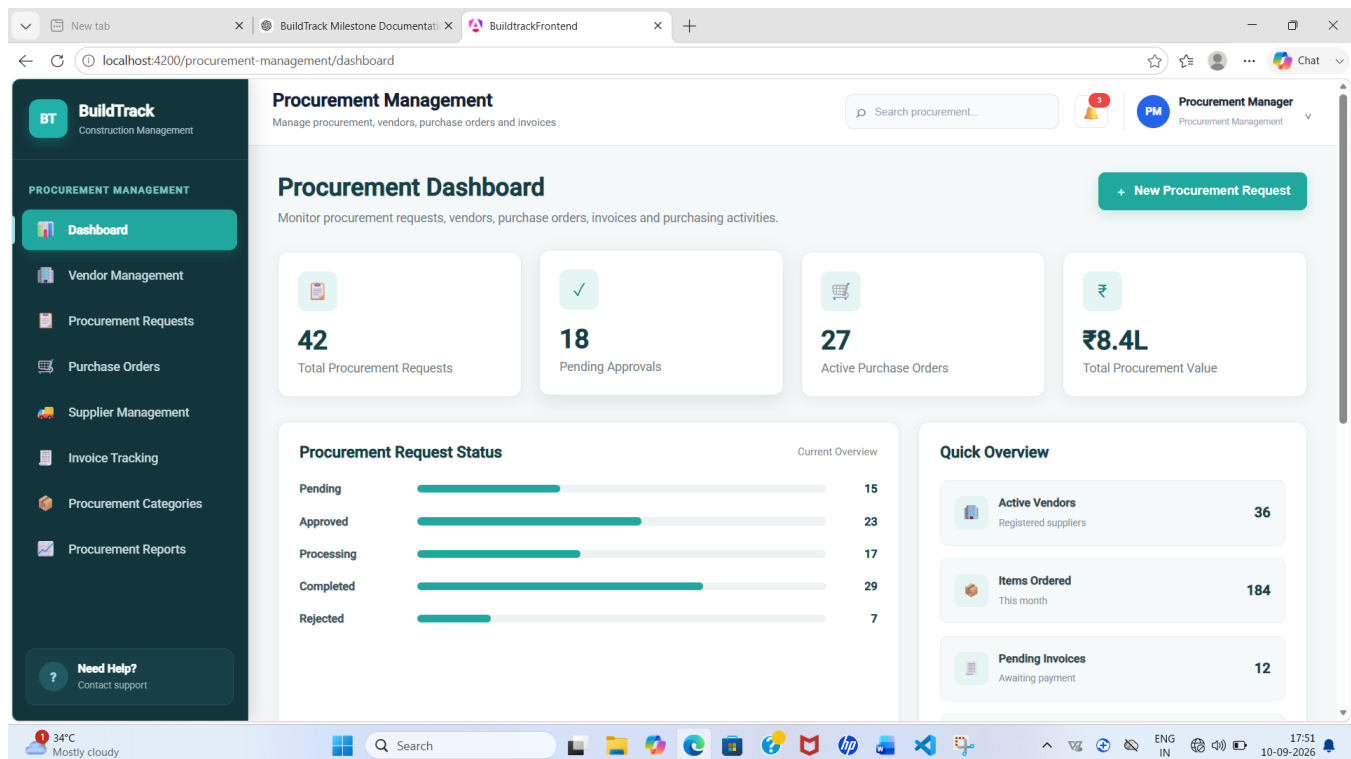


Figure 10: BuildTrack – Procurement Dashboard

6.3 Procurement Requests

Purpose: Create, search and review procurement requests.

How it works in BuildTrack: The page includes a searchable request table with Request ID, Project, Requested By, Item/Material, Category, Quantity, Required Date, Priority, Request Date and Status. A Request Details area captures project, requester, item/material, quantity, required date, priority, purpose/requirement and remarks.

Role/User: Procurement Management user

Expected output: Procurement requests can be entered and reviewed through the interface.

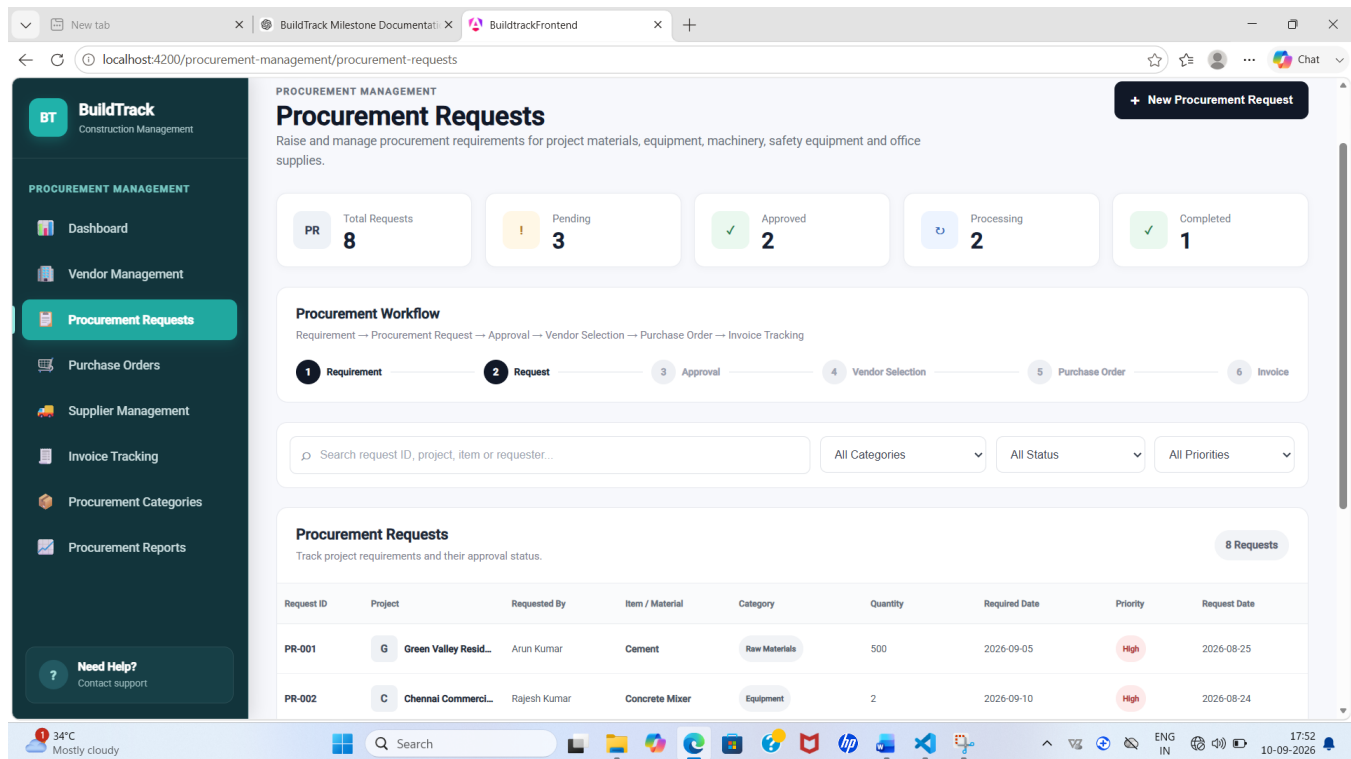


Figure 11: BuildTrack – Procurement Requests

6.4 Purchase Orders

Purpose: Create and monitor purchase orders.

How it works in BuildTrack: The page provides search and a purchase-order table with PO ID, Vendor, Project, Request ID, Item, Category, Order Date, Delivery Date, Quantity, Total Amount and Status. The form includes request, item and quantity/amount-related fields.

Role/User: Procurement Management user

Expected output: Purchase-order information can be reviewed and entered.

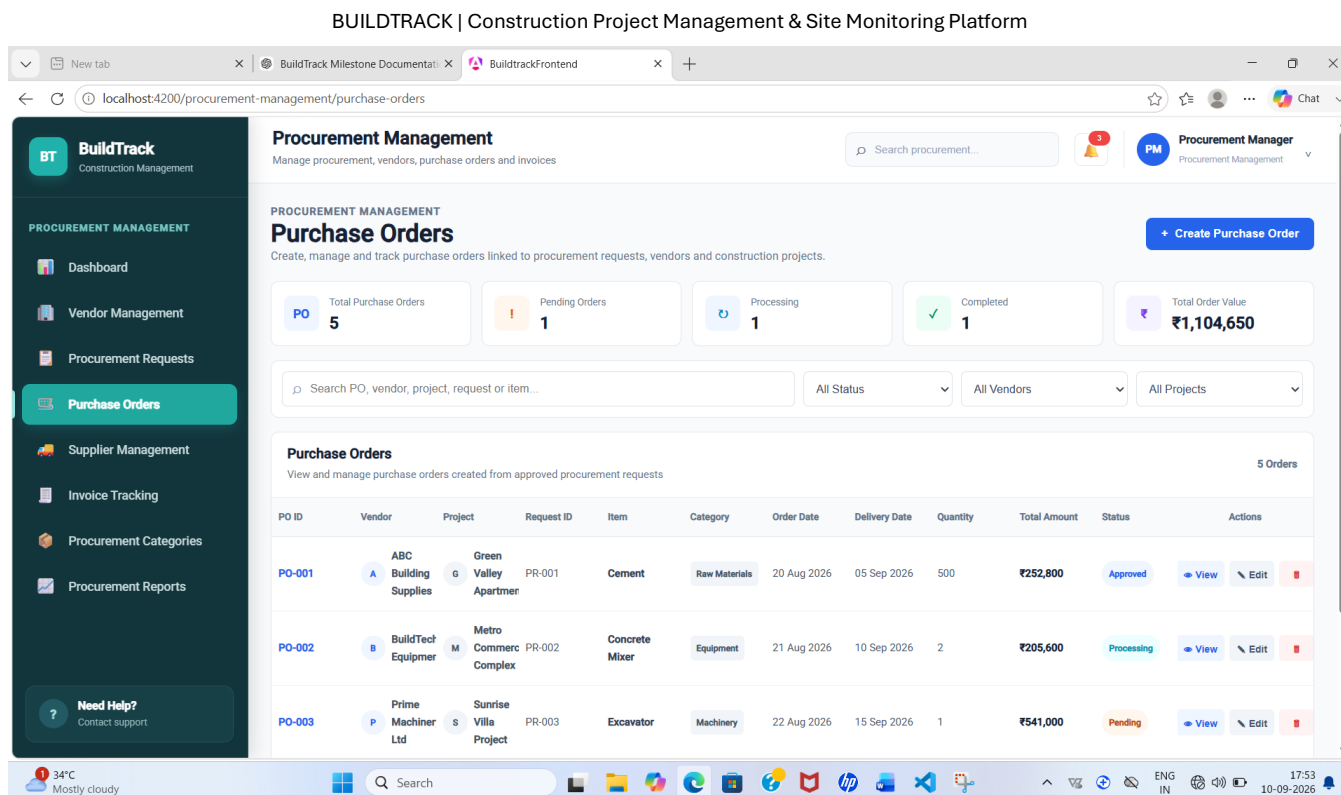


Figure 12: BuildTrack – Purchase Orders

6.5 Invoice Tracking

Purpose: Track vendor/supplier invoices and payment status.

How it works in BuildTrack: The page provides search by invoice, vendor, PO or project and an Invoice Records table containing Invoice ID, Invoice Number, Vendor/Supplier, Purchase Order, Project, Invoice Date, Due Date, Amount, Payment Status and Invoice Status. An Add Invoice form is available.

Role/User: Procurement Management user

Expected output: Invoice information and payment status can be monitored.

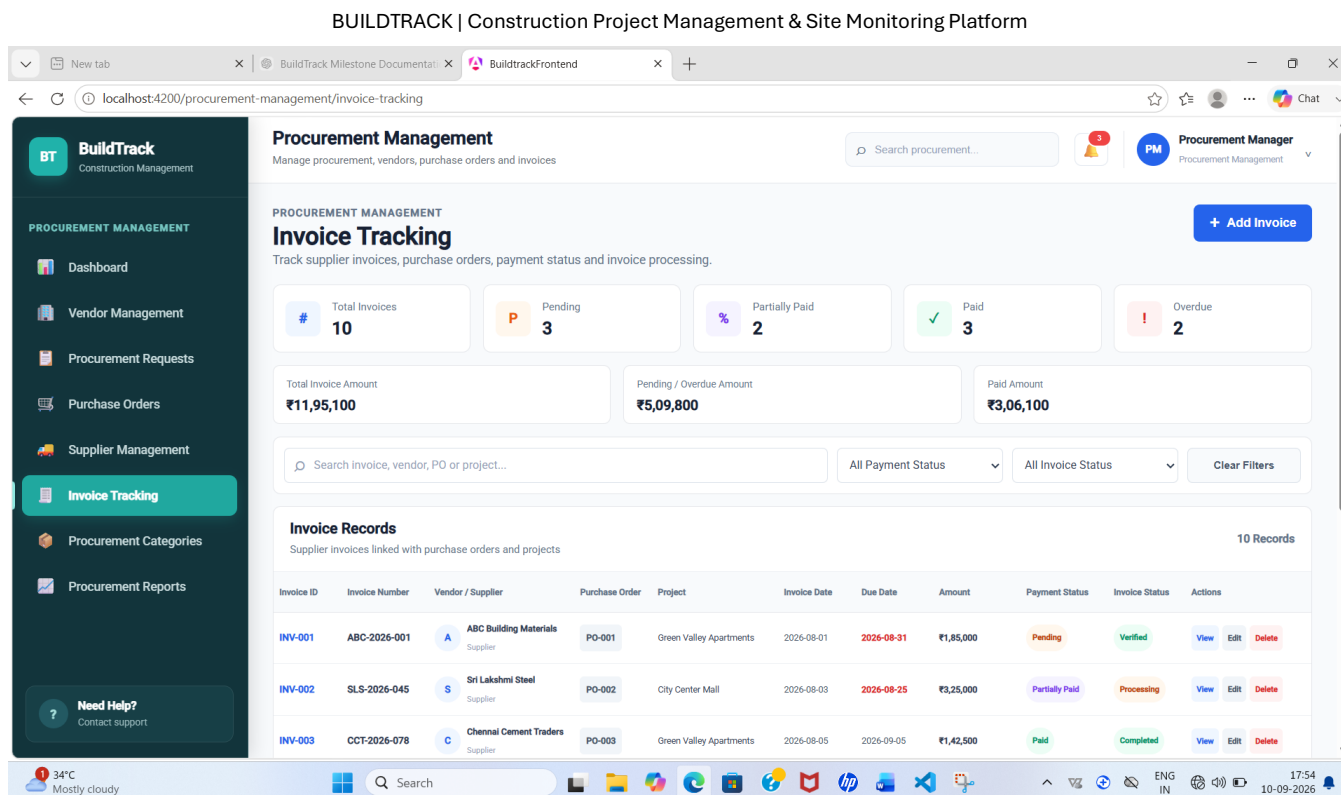


Figure 13: BuildTrack – Invoice Tracking

6.6 Vendor Management

Purpose: Maintain registered vendor information.

How it works in BuildTrack: The Vendor Management page provides search, a Registered Vendors table and Vendor Details. The form includes vendor name, contact person, contact number, email, category, products/services, status and address.

Role/User: Procurement Management user

Expected output: Vendor information can be reviewed and entered.

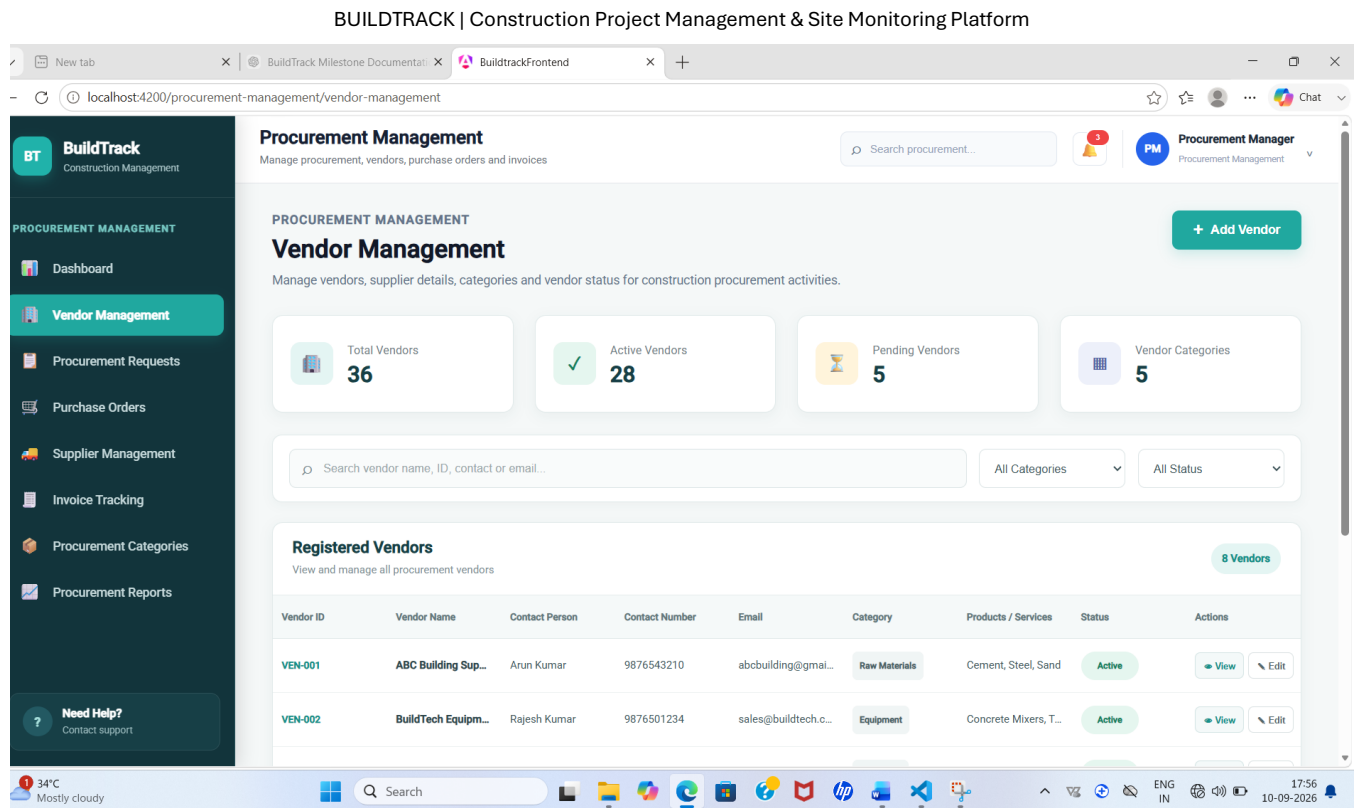


Figure 14: BuildTrack – Vendor Management

6.7 Supplier Management

Purpose: Maintain supplier information relevant to procurement.

How it works in BuildTrack: The Supplier Management page includes search and a Supplier Directory with supplier ID, supplier, contact, category, material/service, projects, rating and status. Supplier details include name, company, contact, phone, email, category, material/service, rating, projects supplied, address and notes.

Role/User: Procurement Management user

Expected output: Supplier information can be reviewed and managed through the UI.

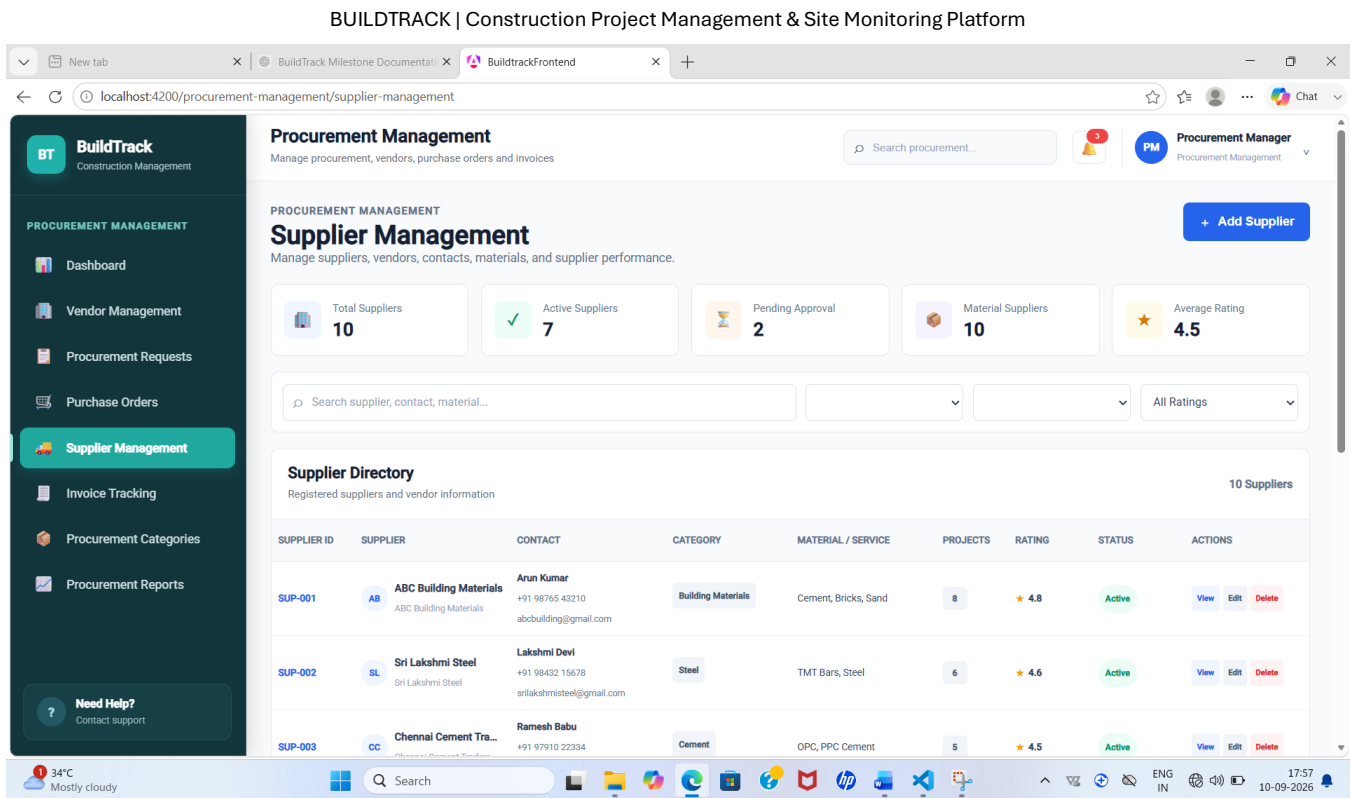


Figure 15: BuildTrack – Supplier Management

6.8 Procurement Categories

Purpose: Organize procurement information by category.

How it works in BuildTrack: The Procurement Categories page provides a category-management interface for procurement classification.

Role/User: Procurement Management user

Expected output: Procurement category information is presented.

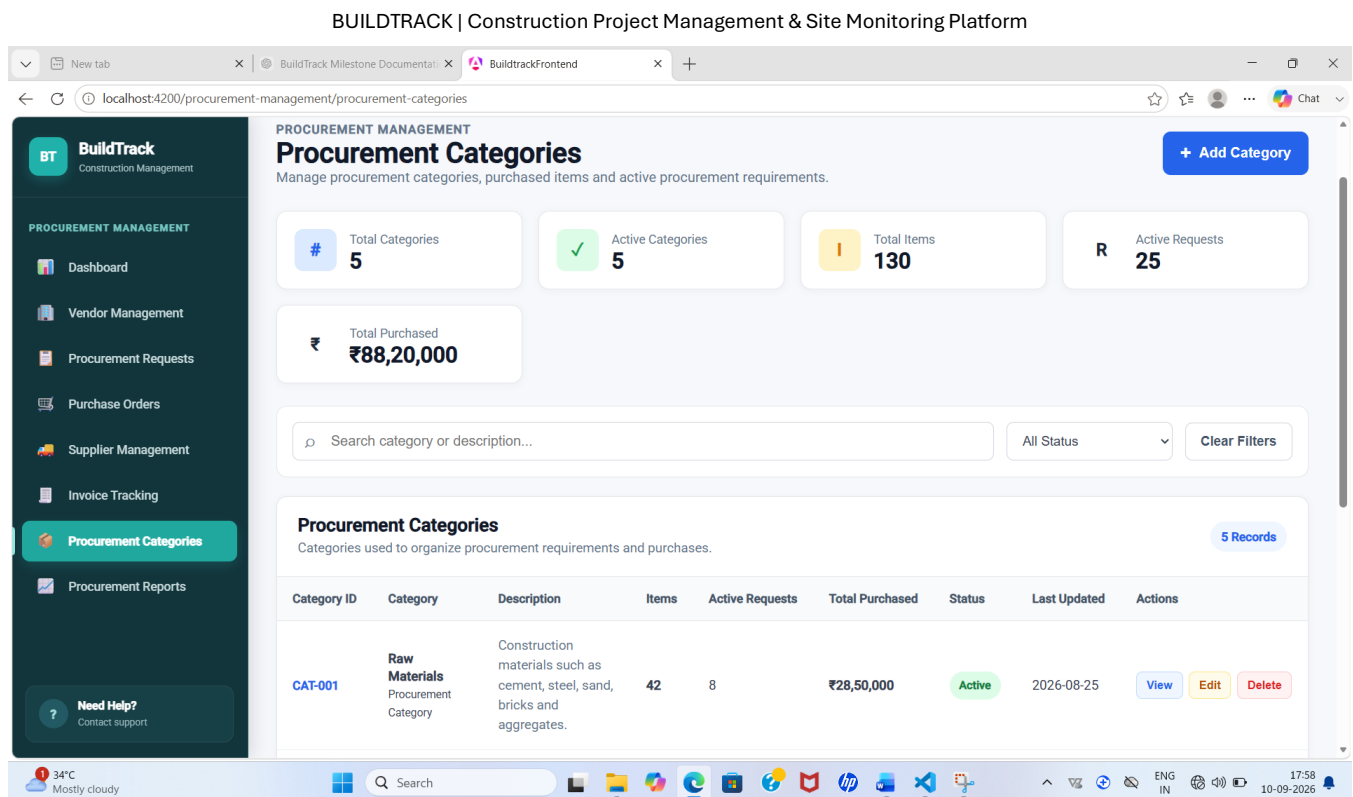


Figure 16: BuildTrack – Procurement Categories

6.9 Procurement Reports

Purpose: Provide procurement reporting information.

How it works in BuildTrack: The Procurement Reports page provides a dedicated procurement-reporting interface.

Role/User: Procurement Management user

Expected output: Procurement reporting information can be accessed.

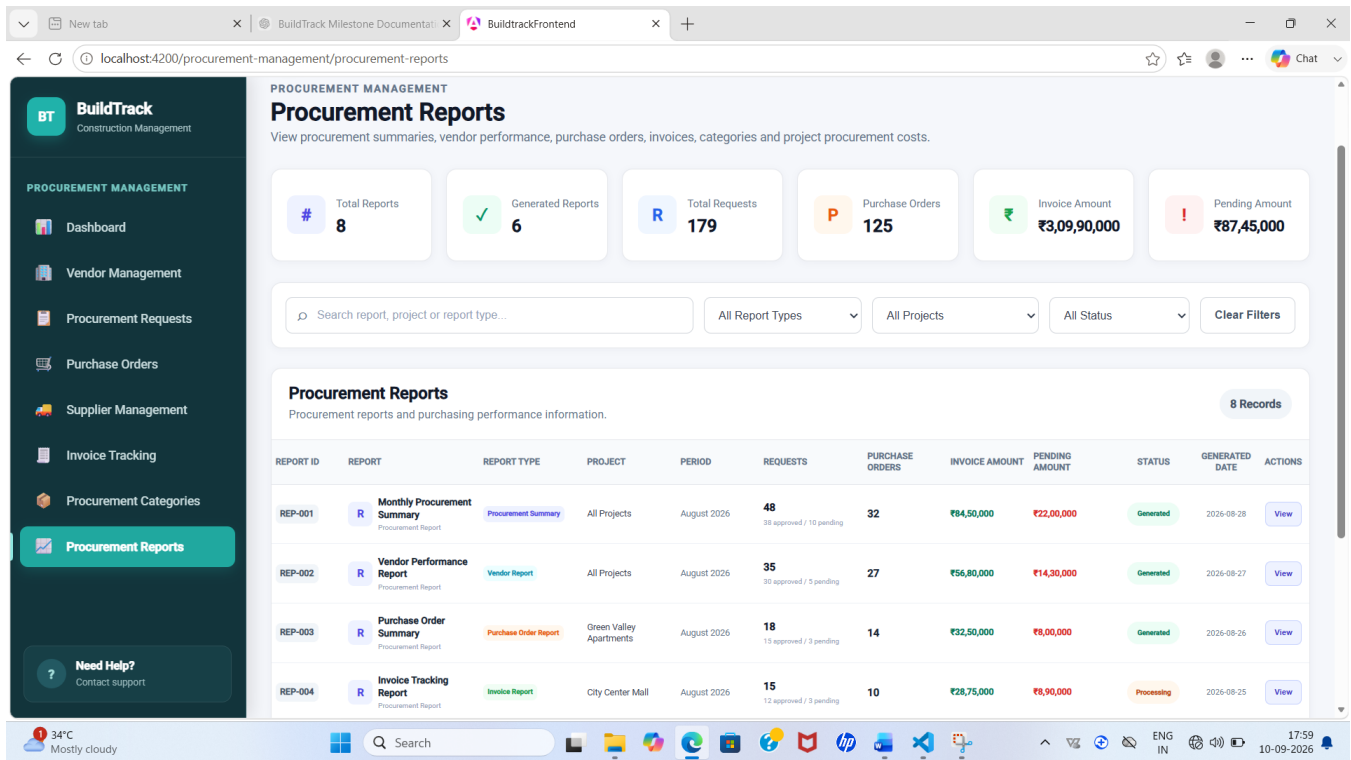


Figure 17: BuildTrack – Procurement Reports

End-to-End Workflow

Step	User Action	System Response	Result
1	Open Procurement Dashboard	Procurement overview is displayed.	Central procurement workspace is available.
2	Create/review a procurement request	Request form/table captures project, requester, item, quantity, dates, priority and status.	Procurement requirement is recorded/reviewed.
3	Review vendors/suppliers	Vendor and Supplier Management pages provide searchable directories and detail forms.	Supplier source information is available.
4	Create/review purchase order	Purchase Orders displays order, vendor, project, item, quantity, amount and status.	Purchase-order information is available.
5	Track invoice	Invoice Tracking displays invoice, PO, project, amount and payment/status fields.	Invoice and payment status can be monitored.
6	Open Procurement Reports	Procurement reporting interface is displayed.	Procurement information can be reviewed in report form.

Module Outcome

The Procurement Management module provides dedicated interfaces for procurement requests, vendors, suppliers, purchase orders, invoice tracking, categories and reporting.

7. Role-wise Functionality

Role / User Area	Milestone 3 Capabilities Evidenced
Material Inventory	Dedicated Admin Material Inventory page exists; complete inventory operations are not claimed because the inspected page is a placeholder.
Workforce Management User	Dashboard, worker registration, attendance tracking, workforce allocation, shift scheduling, payroll monitoring, workforce categories and workforce reports.
Procurement Management User	Dashboard, procurement requests, purchase orders, invoice tracking, vendor management, supplier management, procurement categories and procurement reports.
Other Roles	No additional Milestone 3 capability is claimed without direct evidence from the current source.

8. End-to-End Milestone 3 Workflow

Input	Processing / Action	System View	Output
Material inventory requirement	Open Material Inventory area	Material Inventory page	Inventory entry point; complete workflow pending.
Worker information	Register/review worker	Worker Registration	Worker information available for workforce operations.
Attendance information	Review attendance records	Attendance Tracking	Attendance status and records.
Project workforce requirement	Allocate workers and schedule shifts	Workforce Allocation / Shift Scheduling	Workforce assignment and shift information.
Payroll information	Review payroll records and status	Payroll Monitoring	Payroll monitoring view.
Procurement requirement	Create/review request	Procurement Requests	Procurement requirement information.
Vendor/supplier information	Review or enter source details	Vendor / Supplier Management	Procurement source information.
Purchase requirement	Create/review purchase order	Purchase Orders	PO information and status.
Invoice information	Record/review invoice	Invoice Tracking	Invoice and payment-status information.

Screenshots and Evidence Checklist

Use screenshots from the latest running BuildTrack application. Insert each screenshot immediately after the related feature description, keeping image width, caption style and figure numbering consistent.

- Figure 1 – Material Inventory Page
- Figure 2 – Workforce Dashboard
- Figure 3 – Worker Registration
- Figure 4 – Attendance Tracking
- Figure 5 – Workforce Allocation
- Figure 6 – Shift Scheduling

- Figure 7 – Payroll Monitoring
- Figure 8 – Workforce Categories
- Figure 9 – Workforce Reports
- Figure 10 – Procurement Dashboard
- Figure 11 – Procurement Requests
- Figure 12 – Purchase Orders
- Figure 13 – Invoice Tracking
- Figure 14 – Vendor Management
- Figure 15 – Supplier Management
- Figure 16 – Procurement Categories
- Figure 17 – Procurement Reports

Technical Implementation Summary

Area	Implementation Evidence
Frontend Framework	Angular project structure with standalone components and route-based pages.
UI	Angular templates, forms, data binding, tables, search fields, dashboard summaries and management interfaces.
Routing	Dedicated pages exist for workforce-management and procurement-management areas; Material Inventory is represented by an Admin Material Inventory page.
Backend	Python FastAPI application with routers and database session integration.
Database	SQLAlchemy engine/session configuration and ORM/database layer.
Integration Note	Frontend screens are not treated as proof of live database persistence unless the corresponding API/persistence is demonstrated.

Limitations / Pending Work

- The current Material Inventory page is a placeholder; the reference features of material procurement, inventory monitoring, material requests, material allocation and stock management are not claimed as fully implemented.
- Some Workforce and Procurement screens contain frontend forms/data structures; exact backend persistence should be verified feature by feature.
- The existence of a route or screen alone is not treated as proof of complete API/database integration.
- Final screenshots should be taken from the latest running BuildTrack version.
- Any remaining integration, validation or persistence work should be recorded as pending rather than presented as fully operational.

Milestone Completion Summary

Item	Status / Evidence
Milestone	Milestone 3
Modules Covered	Module 5 – Material Inventory; Module 6 – Workforce Management; Module 7 – Procurement Management
Material Inventory	Dedicated page present; full inventory workflow pending/not sufficiently evidenced.
Workforce Management	Dashboard, registration, attendance, allocation, shifts, payroll, categories and reports present.
Procurement Management	Dashboard, requests, purchase orders, invoice tracking, vendor/supplier management, categories and reports present.
Screenshots	Recommended screenshot positions are provided throughout the document.
Technical Foundation	Angular frontend, FastAPI backend and SQLAlchemy/database layer are evidenced in the project.
Overall Status	Frontend functionality is documented according to source evidence; backend persistence and end-to-end integration require verification where not demonstrated.

9. Conclusion

Milestone 3 extends BuildTrack into workforce and procurement operations while documenting the current state of material inventory functionality. The Workforce Management module provides structured interfaces for worker registration, attendance, allocation, shifts, payroll, categories and reports. The Procurement Management module provides request, vendor, supplier, purchase-order, invoice, category and reporting interfaces. The Material Inventory area is documented conservatively as a partial/placeholder implementation so that the submission reflects the actual project source.